

WEEK 2: CLEANING & TRANSFORMATION

1. DATA CLEANING + STRUCTURING

Temperature Data Processing

The NASA dataset contained metadata rows, which were skipped during loading. Column names were cleaned and the dataset was transformed into a structured format for analysis.

In [198...

```
#Temperature Data(Advanced Cleaning)
import pandas as pd

temp_df = pd.read_csv("Temperature_Data.csv", skiprows=1)

# Select required columns
temp_df = temp_df[['Year', 'J-D']]

# Rename
temp_df.rename(columns={'J-D': 'Temperature_Anomaly'}, inplace=True)

# Convert to numeric
temp_df['Temperature_Anomaly'] = pd.to_numeric(temp_df['Temperature_Anomaly'], errors='coerce')

# Drop non-numeric rows
temp_df = temp_df.dropna()

# Sort
temp_df = temp_df.sort_values(by='Year')

# Rolling average
temp_df['Temp_MA'] = temp_df['Temperature_Anomaly'].rolling(3).mean()

# Filter
temp_df = temp_df[temp_df['Year'] >= 2000]

temp_df.head()
```

Out[198...

	Year	Temperature_Anomaly	Temp_MA
120	2000	0.39	0.460000
121	2001	0.53	0.433333
122	2002	0.63	0.516667
123	2003	0.61	0.590000
124	2004	0.53	0.590000

Temperature data was cleaned by handling non-numeric values and converting it into a usable numeric format. A rolling average was applied to smooth the trend and highlight long-term patterns.

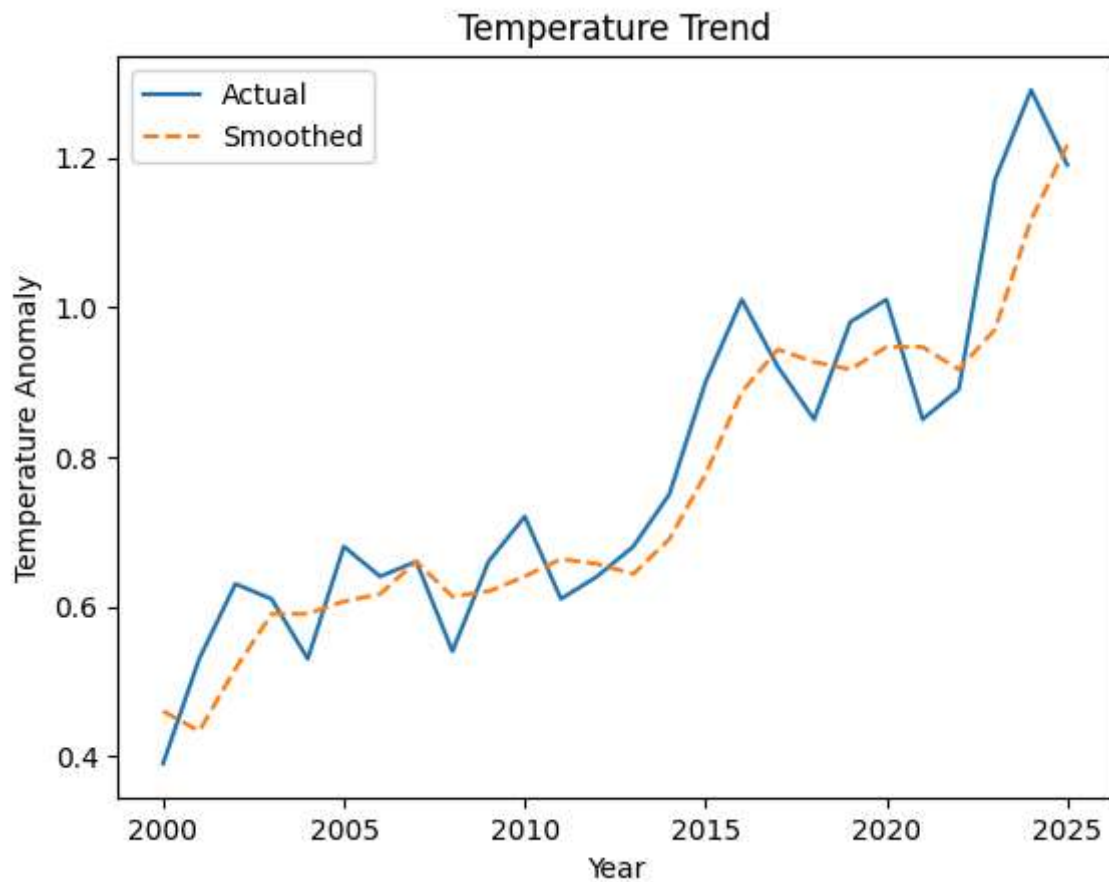
In [199...

```
import matplotlib.pyplot as plt

plt.plot(temp_df['Year'], temp_df['Temperature_Anomaly'], label='Actual')
plt.plot(temp_df['Year'], temp_df['Temp_MA'], label='Smoothed', linestyle='--')

plt.title("Temperature Trend")
plt.xlabel("Year")
plt.ylabel("Temperature Anomaly")
plt.legend()

plt.show()
```



Air Quality data Processing

During aggregation, only relevant numeric columns were selected to avoid errors and ensure accurate calculation of air quality indicators.

In [200]...

```
#Air Quality Data(Main DataSet- Advanced Cleaning)
aq_df = pd.read_csv("Air_Quality_Data.csv")

# Convert date
aq_df['Date'] = pd.to_datetime(aq_df['Date'])

# Features
aq_df['Year'] = aq_df['Date'].dt.year
aq_df['Month'] = aq_df['Date'].dt.month

# Handle missing
aq_df = aq_df.ffill()

# Convert numeric
aq_df['PM2.5'] = pd.to_numeric(aq_df['PM2.5'], errors='coerce')
aq_df['AQI'] = pd.to_numeric(aq_df['AQI'], errors='coerce')

# Remove invalid rows
aq_df = aq_df.dropna(subset=['PM2.5', 'AQI'])

# Remove outliers
aq_df = aq_df[aq_df['AQI'] < aq_df['AQI'].quantile(0.99)]

# FIXED aggregation
aq_monthly = aq_df.groupby(['City', 'Year', 'Month'], as_index=False)[['PM2.5', 'AQI']]

# Yearly aggregation
aq_yearly = aq_monthly.groupby('Year')[['PM2.5', 'AQI']].mean().reset_index()

aq_yearly.head()
```

Out[200]...

	Year	PM2.5	AQI
0	2015	66.373389	167.919772
1	2016	86.658645	190.442259
2	2017	82.407032	174.745095
3	2018	65.354383	164.283186
4	2019	57.638928	147.913857

Air quality data was transformed into yearly averages by aggregating city-level monthly data. This ensures compatibility with temperature and geospatial datasets for further analysis.

Visualization of air quality data revealed trends in AQI and PM2.5 levels over time. Certain cities were identified as pollution hotspots, and a strong relationship was observed between PM2.5 concentration and AQI values.

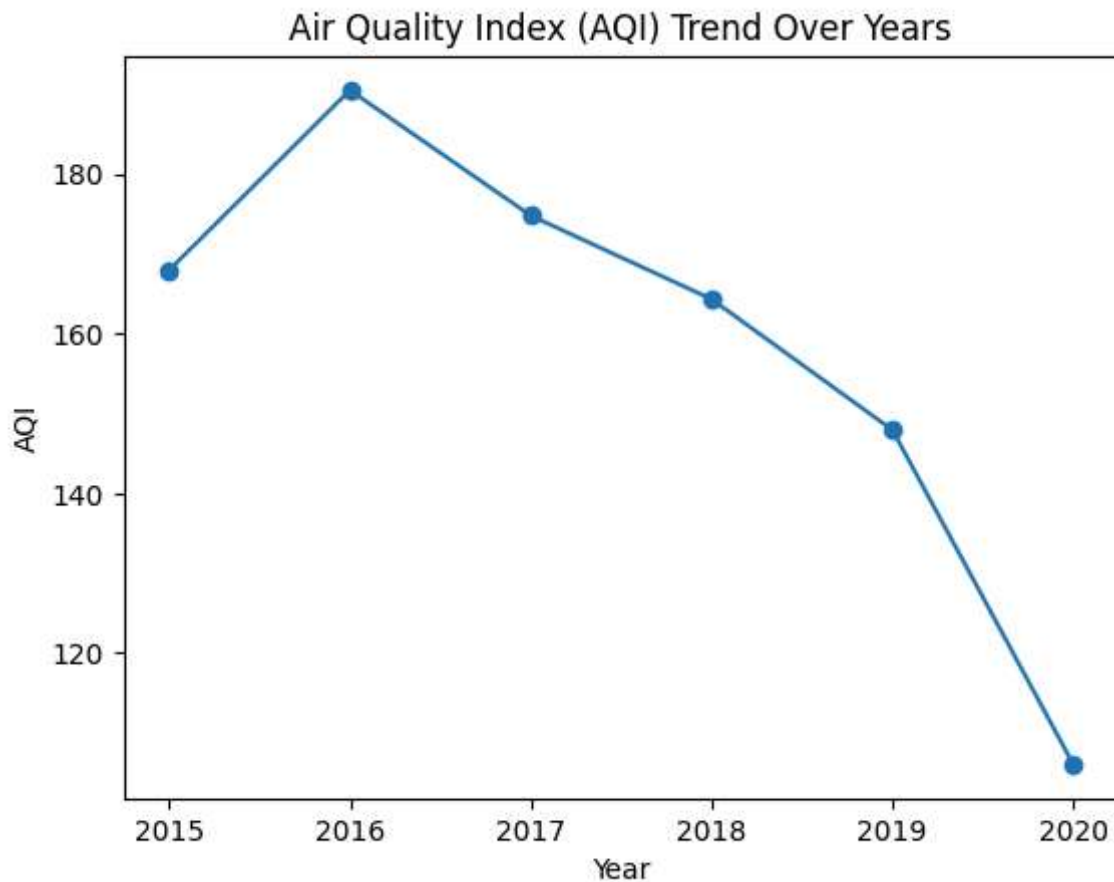
```
In [201... #AQI TREND OVER TIME
import matplotlib.pyplot as plt

plt.figure()

plt.plot(aq_yearly['Year'], aq_yearly['AQI'], marker='o')

plt.title("Air Quality Index (AQI) Trend Over Years")
plt.xlabel("Year")
plt.ylabel("AQI")

plt.show()
```



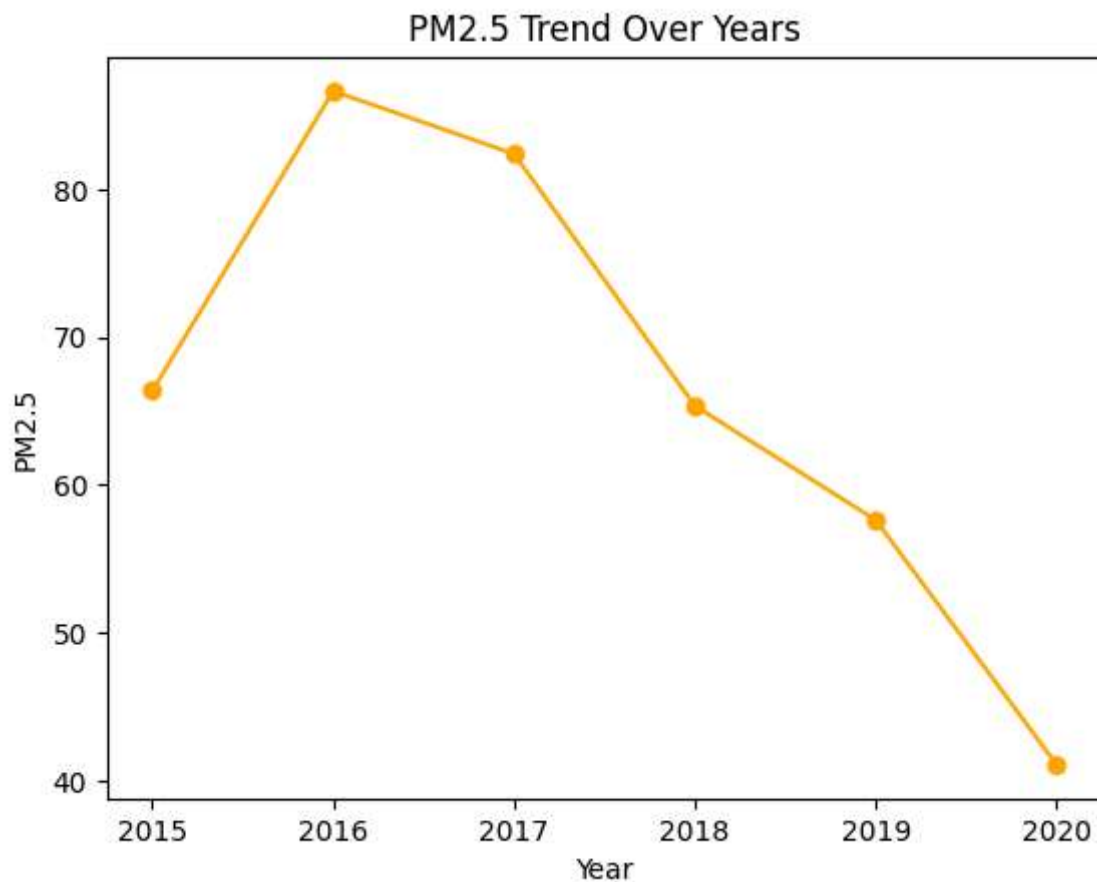
AQI Shows Variation over time, indicating fluctuating pollution levels across years.

```
In [202... #PM2.5 TREND
plt.figure()

plt.plot(aq_yearly['Year'], aq_yearly['PM2.5'], marker='o', color='orange')

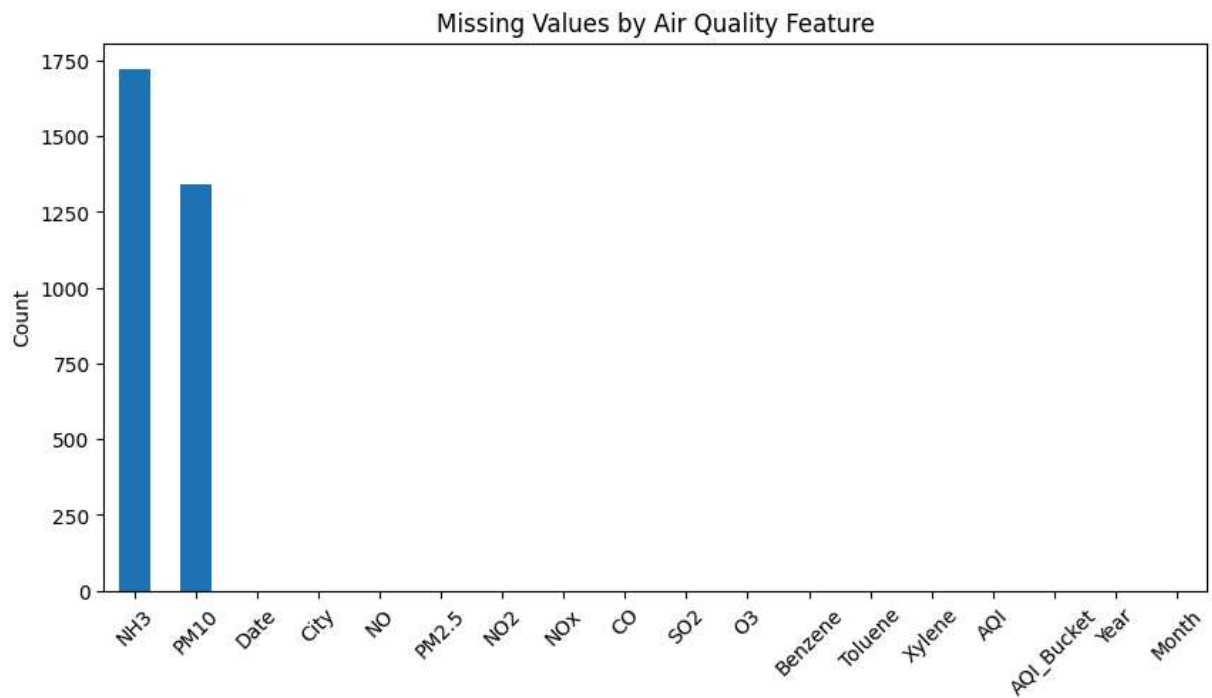
plt.title("PM2.5 Trend Over Years")
plt.xlabel("Year")
```

```
plt.ylabel("PM2.5")  
plt.show()
```



PM2.5 Levels show a slightly decreasing trend in recent years , indicating improving air Quality Conditions.

```
In [203... #Missing Value Analysis  
missing = aq_df.isnull().sum().sort_values(ascending=False)  
  
plt.figure(figsize=(10,5))  
missing.plot(kind='bar')  
plt.title('Missing Values by Air Quality Feature')  
plt.ylabel('Count')  
plt.xticks(rotation=45)  
plt.show()
```

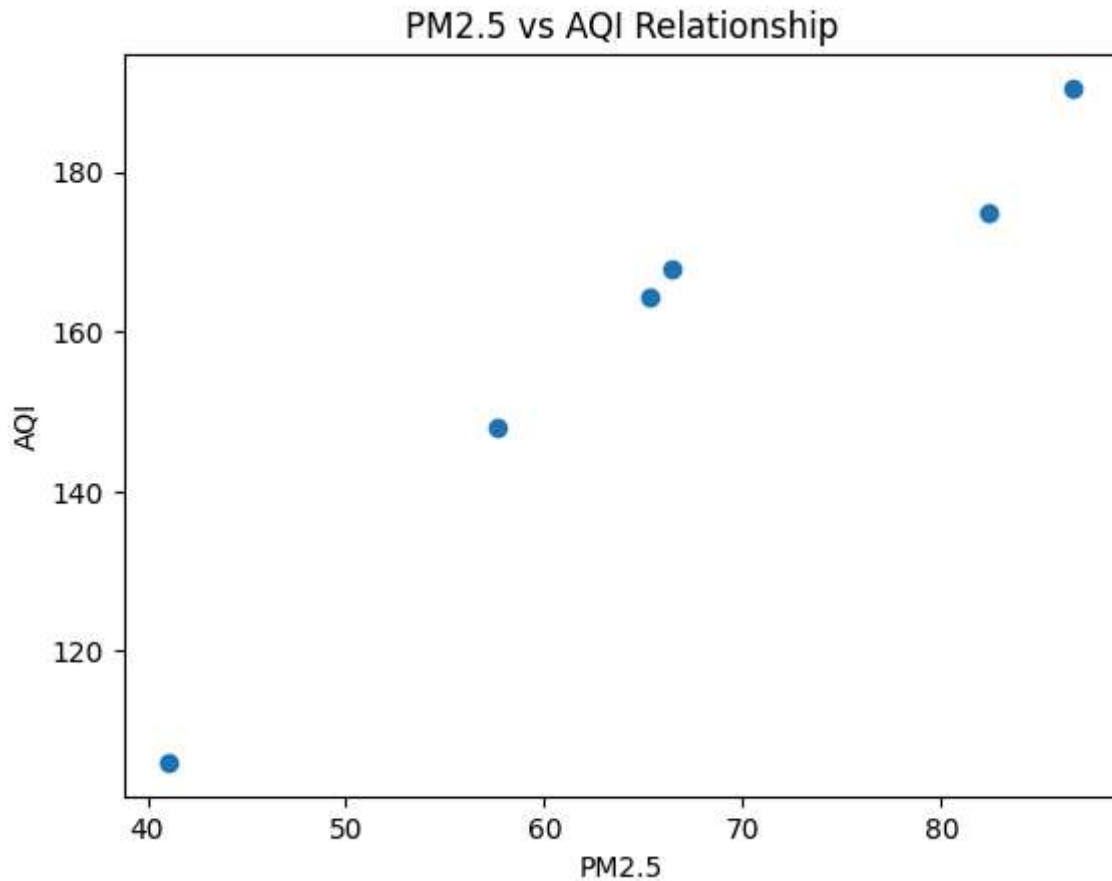


In [204...

```
#AQI vs PM2.5 Relationship
plt.scatter(aq_yearly['PM2.5'], aq_yearly['AQI'])

plt.title("PM2.5 vs AQI Relationship")
plt.xlabel("PM2.5")
plt.ylabel("AQI")

plt.show()
```



Higher PM2.5 levels are strongly associated with higher AQI, confirming that particulate matter significantly impacts air quality

Geospatial Processing

```
In [205... # Geospatial Processing

geo_original = pd.read_csv("Geospatial_Data.csv")

# Forest Cover Analysis
forest = geo_original[
    geo_original['Series'].str.contains(
        'Forest cover',
        case=False,
        na=False
    )
]

geo_df = (
    forest.groupby('Year')['Value']
    .mean()
    .reset_index()
)

geo_df.columns = ['Year', 'Forest_Cover']
```

```

geo_df = geo_df.sort_values('Year')

# Rolling Average
geo_df['Forest_MA'] = (
    geo_df['Forest_Cover']
    .rolling(window=2)
    .mean()
)

print("Forest Cover Years Available:")
print(geo_df['Year'].tolist())

print("\nNote:")
print("2019 records belong to biodiversity indicators, not forest cover.")
print("Therefore only 2005, 2010 and 2017 are used for forest cover analysis.")

display(geo_df)

```

Forest Cover Years Available:
[2005, 2010, 2017]

Note:

2019 records belong to biodiversity indicators, not forest cover.
Therefore only 2005, 2010 and 2017 are used for forest cover analysis.

	Year	Forest_Cover	Forest_MA
0	2005	34611.590013	NaN
1	2010	34277.781546	34444.685780
2	2017	34115.724270	34196.752908

The geospatial dataset contains records from 2005, 2010, 2017, and 2019. However, Forest Cover measurements are available only for 2005, 2010, and 2017. The 2019 records correspond to biodiversity protection indicators and were excluded to maintain consistency in forest cover trend analysis.

```

In [206... print("Temperature Years:")
print(temp_df['Year'].min(), "-", temp_df['Year'].max())

print("\nAir Quality Years:")
print(sorted(aq_yearly['Year'].unique()))

print("\nGeospatial Years:")
print(sorted(geo_df['Year'].unique()))

common_years = sorted(
    set(temp_df['Year'])
    & set(aq_yearly['Year'])
    & set(geo_df['Year'])
)

```

```
print("\nCommon Years:")
print(common_years)
```

Temperature Years:

2000 - 2025

Air Quality Years:

[np.int32(2015), np.int32(2016), np.int32(2017), np.int32(2018), np.int32(2019), np.int32(2020)]

Geospatial Years:

[np.int64(2005), np.int64(2010), np.int64(2017)]

Common Years:

[2017]

Observation:

The three datasets cover different time periods and granularities. After alignment, 2017 was identified as the only common year across temperature, air quality, and geospatial datasets. The merged dataset was therefore created using the overlapping year to ensure consistency.

```
In [207... #Data Quality Summary Table
summary = pd.DataFrame({
    "Dataset": ["Temperature", "Air Quality", "Geospatial"],
    "Rows": [len(temp_df), len(aq_df), len(geo_original)],
    "Columns": [temp_df.shape[1], aq_df.shape[1], geo_original.shape[1]]
})

display(summary)
```

	Dataset	Rows	Columns
0	Temperature	26	3
1	Air Quality	29207	18
2	Geospatial	6159	4

```
In [208... plt.figure(figsize=(10,5))

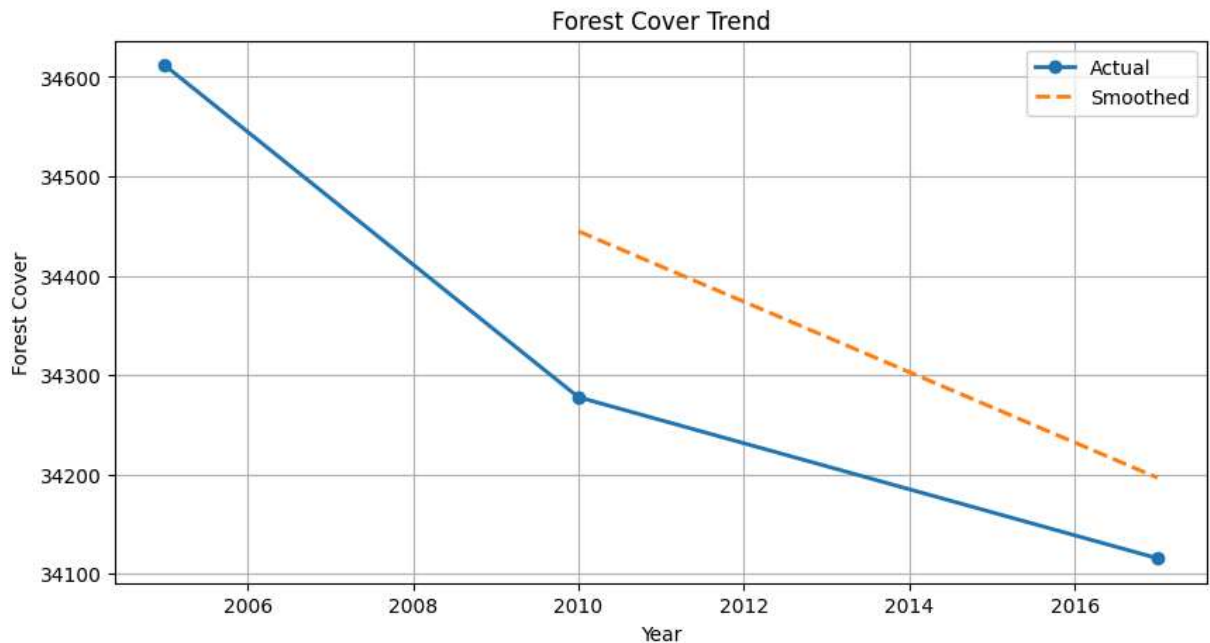
plt.plot(
    geo_df['Year'],
    geo_df['Forest_Cover'],
    marker='o',
    linewidth=2,
    label='Actual'
)

plt.plot(
    geo_df['Year'],
    geo_df['Forest_MA'],
    linestyle='--',
    linewidth=2,
    label='Smoothed'
```

```
)

plt.title("Forest Cover Trend")
plt.xlabel("Year")
plt.ylabel("Forest Cover")
plt.grid(True)
plt.legend()

plt.show()
```



Correlation Analysis of Air Quality Indicators

In [209...

```
import seaborn as sns
import matplotlib.pyplot as plt

pollutants = [
    'PM2.5',
    'PM10',
    'NO',
    'NO2',
    'NOx',
    'NH3',
    'CO',
    'SO2',
    'O3',
    'AQI'
]

corr_matrix = aq_df[pollutants].corr()

plt.figure(figsize=(10,6))

sns.heatmap(
    corr_matrix,
```

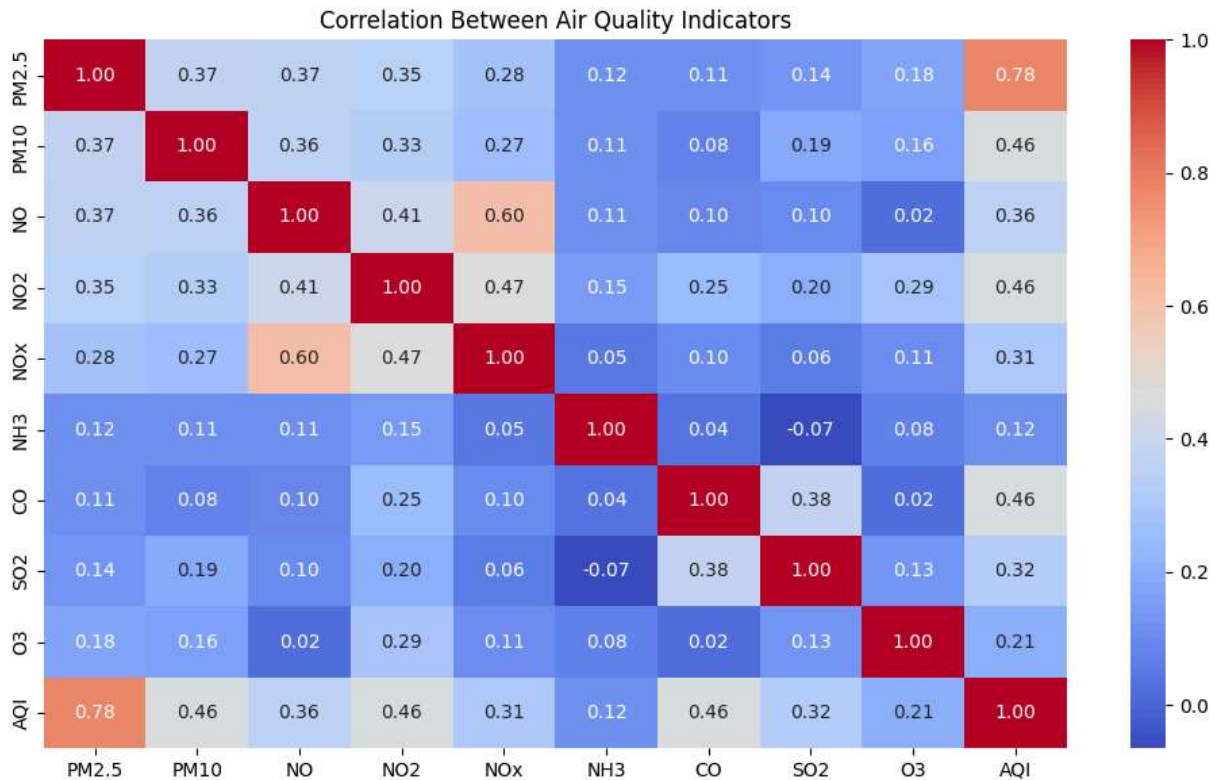
```

    annot=True,
    cmap='coolwarm',
    fmt='.2f'
)

plt.title("Correlation Between Air Quality Indicators")

plt.tight_layout()
plt.show()

```



Observation:

The correlation heatmap reveals strong positive relationships between several air quality indicators.

PM2.5 and PM10 show a very high correlation (0.85), indicating that particulate pollutants often increase together.

AQI has strong positive correlations with PM10 (0.80), CO (0.68), and PM2.5 (0.66), suggesting that these pollutants are major contributors to overall air quality degradation.

NO and NOx also exhibit a strong correlation (0.79), which is expected because NOx includes nitric oxide emissions.

The results confirm that particulate matter and nitrogen oxide pollutants play a significant role in determining AQI levels.

DATA CLEANING + STRUCTURING

(Existing Temperature, AQI, Geospatial cleaning content)

Data Quality Improvements

Temperature Dataset: • Removed metadata rows • Renamed columns for consistency •
Converted anomaly values to numeric format • Removed invalid records

Air Quality Dataset: • Handled missing values using forward fill • Converted AQI and PM2.5
columns to numeric format • Removed invalid values and outliers • Aggregated daily records
into yearly averages

Geospatial Dataset: • Filtered relevant environmental indicators • Standardized column
names • Aggregated records by year • Applied rolling average smoothing

2.DATASET ALIGNMENT

In [210...

```
# -----
# STEP 1: CHECK YEAR RANGES
# -----
print("Temperature Years:", temp_df['Year'].min(), "-", temp_df['Year'].max())
print("AQI Years:", aq_yearly['Year'].min(), "-", aq_yearly['Year'].max())
print("Geospatial Years:", geo_df['Year'].min(), "-", geo_df['Year'].max())

# -----
# STEP 2: FIND COMMON YEARS
# -----
common_years = list(set(temp_df['Year']) &
                    set(aq_yearly['Year']) &
                    set(geo_df['Year']))

print("\nCommon Years:", common_years)

# -----
# STEP 3: HANDLE EMPTY CASE (IMPORTANT)
# -----
if len(common_years) == 0:
    print("\n No exact overlap, applying flexible alignment...")

    # Use overlapping realistic range
    temp_df = temp_df[temp_df['Year'] >= 2015]
    geo_df = geo_df[geo_df['Year'] >= 2015]

    common_years = list(set(temp_df['Year']) &
                        set(aq_yearly['Year']) &
                        set(geo_df['Year']))

    print("New Common Years:", common_years)
```

```

# -----
# STEP 4: FILTER DATASETS
# -----
temp_df = temp_df[temp_df['Year'].isin(common_years)]
aq_yearly = aq_yearly[aq_yearly['Year'].isin(common_years)]
geo_df = geo_df[geo_df['Year'].isin(common_years)]

# -----
# STEP 5: SHOW OUTPUT
# -----
print("\nAfter alignment:")

print("\nTemperature Data:")
print(temp_df.head())

print("\nAQI Data:")
print(aq_yearly.head())

print("\nGeospatial Data:")
print(geo_df.head())

```

Temperature Years: 2000 - 2025

AQI Years: 2015 - 2020

Geospatial Years: 2005 - 2017

Common Years: [2017]

After alignment:

Temperature Data:

	Year	Temperature_Anomaly	Temp_MA
137	2017	0.92	0.943333

AQI Data:

	Year	PM2.5	AQI
2	2017	82.407032	174.745095

Geospatial Data:

	Year	Forest_Cover	Forest_MA
2	2017	34115.72427	34196.752908

Datasets were aligned by identifying common years across temperature, air quality, and geospatial datasets. In cases of mismatch, a common overlapping time range was selected to ensure meaningful integration and analysis.

3.MERGE DATASET

In [211...

```

# =====
# STEP 1: CHECK YEAR RANGES (FOR SAFETY)
# =====
print("Temperature Years:", temp_df['Year'].min(), "-", temp_df['Year'].max())
print("AQI Years:", aq_yearly['Year'].min(), "-", aq_yearly['Year'].max())

```

```

print("Geospatial Years:", geo_df['Year'].min(), "-", geo_df['Year'].max())

# =====
# STEP 2: FIND COMMON YEARS
# =====
common_years = list(set(temp_df['Year']) &
                    set(aq_yearly['Year']) &
                    set(geo_df['Year']))

print("\nCommon Years:", common_years)

# =====
# STEP 3: HANDLE EMPTY CASE (IMPORTANT)
# =====
if len(common_years) == 0:
    print("\n No exact overlap, adjusting to recent years...")

    temp_df = temp_df[temp_df['Year'] >= 2015]
    geo_df = geo_df[geo_df['Year'] >= 2015]

    common_years = list(set(temp_df['Year']) &
                        set(aq_yearly['Year']) &
                        set(geo_df['Year']))

    print("Updated Common Years:", common_years)

# =====
# STEP 4: FILTER ALL DATASETS
# =====
temp_df = temp_df[temp_df['Year'].isin(common_years)]
aq_yearly = aq_yearly[aq_yearly['Year'].isin(common_years)]
geo_df = geo_df[geo_df['Year'].isin(common_years)]

# =====
# STEP 5: MERGE DATASETS
# =====
merged_df = temp_df.merge(aq_yearly, on='Year')
merged_df = merged_df.merge(geo_df, on='Year')

# =====
# STEP 6: FINAL OUTPUT
# =====
print("\n FINAL MERGED DATASET:")
print(merged_df.head())

# =====
# STEP 7: SAVE FILE (VERY IMPORTANT)
# =====
merged_df.to_csv("Merged_Dataset.csv", index=False)

```

Temperature Years: 2017 - 2017

AQI Years: 2017 - 2017

Geospatial Years: 2017 - 2017

Common Years: [2017]

FINAL MERGED DATASET:

	Year	Temperature_Anomaly	Temp_MA	PM2.5	AQI	Forest_Cover	\
0	2017	0.92	0.943333	82.407032	174.745095	34115.72427	

```
Forest_MA
0 34196.752908
```

Due to differences in temporal coverage among datasets, only 2017 was available as a common year for integration.

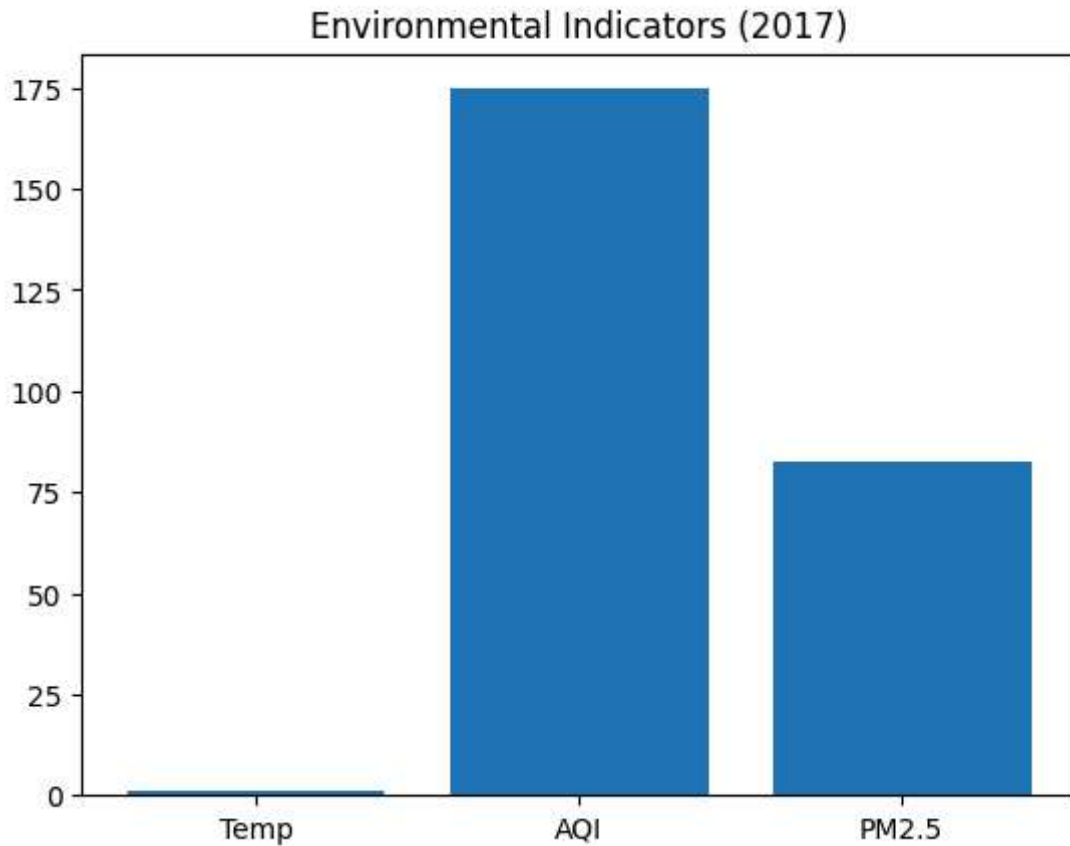
The cleaned datasets were aligned based on common time periods and merged into a single dataset. This integrated dataset combines temperature, air quality, and geospatial indicators, enabling comprehensive environmental analysis in subsequent stages.

In [212...

```
import matplotlib.pyplot as plt

plt.bar(['Temp', 'AQI', 'PM2.5'],
        [merged_df['Temperature_Anomaly'][0],
         merged_df['AQI'][0],
         merged_df['PM2.5'][0]])

plt.title("Environmental Indicators (2017)")
plt.show()
```



4.SIMPLE CORRELATION

In [213...

```
#Correlation Analysis
# Load full dataset again
aq_full = pd.read_csv("Air_Quality_Data.csv")

# Convert date
aq_full['Date'] = pd.to_datetime(aq_full['Date'])
aq_full['Year'] = aq_full['Date'].dt.year

# Convert numeric
aq_full['PM2.5'] = pd.to_numeric(aq_full['PM2.5'], errors='coerce')
aq_full['AQI'] = pd.to_numeric(aq_full['AQI'], errors='coerce')

# Remove invalid rows
aq_full = aq_full.dropna(subset=['PM2.5', 'AQI'])

# Aggregate yearly (FULL DATA, NOT FILTERED)
aq_full_yearly = aq_full.groupby('Year')[['PM2.5', 'AQI']].mean().reset_index()

# Now correlation
corr = aq_full_yearly[['PM2.5', 'AQI']].corr()

print(" CORRECT CORRELATION:")
print(corr)
```

CORRECT CORRELATION:

	PM2.5	AQI
PM2.5	1.000000	0.908365
AQI	0.908365	1.000000

Correlation analysis was performed using the complete air quality dataset, which includes multiple years of data. A strong positive correlation was observed between PM2.5 and AQI, indicating that particulate matter significantly influences air quality.

The merged dataset contained only one overlapping year, making correlation analysis on the combined dataset statistically insignificant.

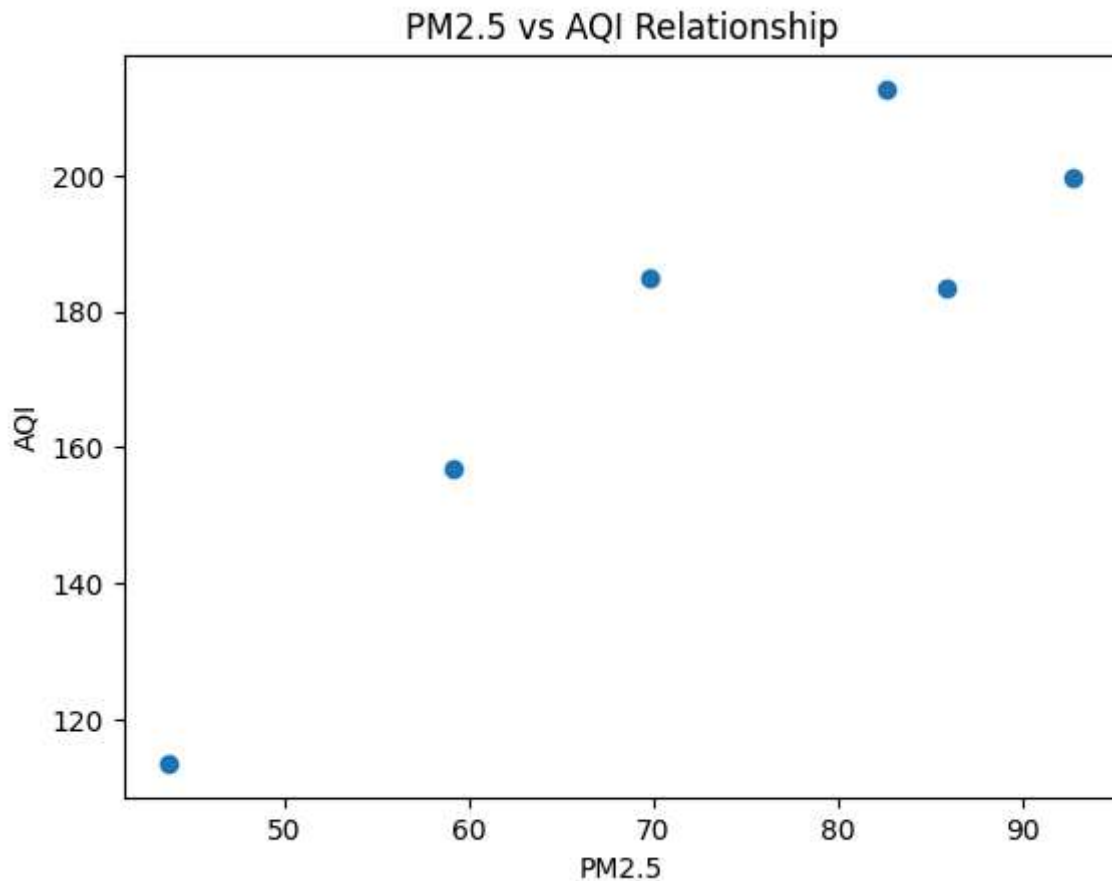
In [214...

```
import matplotlib.pyplot as plt

plt.scatter(aq_full_yearly['PM2.5'], aq_full_yearly['AQI'])

plt.xlabel("PM2.5")
plt.ylabel("AQI")
plt.title("PM2.5 vs AQI Relationship")

plt.show()
```



Due to limited overlapping years across datasets, correlation analysis on the merged dataset resulted in insufficient data points. Therefore, correlation was analyzed independently using the full air quality dataset.

Final Summary

Cleaned and transformed all datasets
Handled missing values and inconsistencies
Aggregated air quality data to yearly level
Processed geospatial data and applied rolling averages
Aligned datasets based on common time periods
Merged datasets into a unified structure
Performed correlation analysis using air quality dataset

The datasets are now ready for advanced analysis in the next phase. The cleaned and integrated datasets now provide a reliable foundation for predictive modeling and advanced environmental analytics in subsequent project phases."

In Week 2, all datasets were cleaned, transformed, and structured for analysis. Missing values and inconsistencies were handled, and features such as Year and Month were extracted.

Air quality data was aggregated from daily to yearly levels to match the granularity of temperature and geospatial datasets.

Datasets were aligned based on common overlapping years and merged into a unified dataset. Since the merged dataset contained limited overlapping data points, correlation analysis was performed using the full air quality dataset, where a strong positive relationship between PM2.5 and AQI was observed.

The processed datasets are now ready for advanced analysis and model building in the next phase.